

Assignment 1 Report

Group - 13

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2nd March 2026

1 Introduction

The results in this report were obtained using a custom Intel PIN tool implemented in `HW1.cpp`. The tool instruments SPEC CPU2006 benchmarks compiled for the ia32 (32-bit x86) architecture and collects statistics such as instruction profiles, CPI, memory footprints, and ISA properties.

All experiments were conducted on a Linux x86 system using Intel PIN 4.1. The PIN root directory was configured as:

```
export PIN_ROOT=~/.pin-external-4.1-99687-gd9b8f822c-gcc-linux
```

Since the provided SPEC binaries are 32-bit, the pintool was compiled for the ia32 target using:

```
cd $PIN_ROOT/source/tools/CS422
make TARGET=ia32 obj-ia32/HW1.so
```

This generates the pintool shared object:

```
$PIN_ROOT/source/tools/CS422/obj-ia32/HW1.so
```

The tool is loaded by the PIN runtime (not executed directly) and invoked using the PIN driver while running the benchmarks. During execution, the tool fast-forwards the benchmark by a specified number of instructions and then collects statistics for the next one billion instructions before terminating.

For submission, the project directory containing `HW1.cpp`, the makefiles, scripts, and the report was archived into a zip file after removing generated build artifacts such as `obj-*`, `.o`, and `.so` files.

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2 400.perlbench

Part 1: Dynamic Instruction Profile

Fast-Forward Count : 207000000000

Total Instructions : 208000000005

#	Instruction Type	Count	Percent (%)
1	Loads	355701101	22.768
2	Stores	206749417	13.234
3	NOPs	993048	0.064
4	Direct Calls	12970842	0.830
5	Indirect Calls	2781696	0.178
6	Returns	15752540	1.008
7	Unconditional Branches	30458025	1.950
8	Conditional Branches	130139652	8.330
9	Logical Operations	100625810	6.441
10	Rotate & Shift	4249337	0.272
11	Flag Operations	876471	0.056
12	Vector (AVX)	0	0.000
13	Conditional Moves	0	0.000
14	MMX/SSE	0	0.000
15	System Calls	0	0.000
16	Floating Point	979720	0.063
17	The Rest	700010229	44.807
Total		1562287888	

Part 2: CPI Analysis

Total Cycles : 40371373630

Total Instructions : 1562287888

CPI : 25.8412

Part 3: Memory Footprint

Unique 32-byte Instruction Chunks : 2812

Unique 32-byte Data Chunks : 31366

Instruction Chunks [single : multi] : 921826989 : 78173009

Data Chunks [single : multi] : 552258993 : 25031

Part 4: ia32 ISA Properties

Instruction Length Distribution

Length (Bytes)	Count
1	117805271
2	257141215
3	275248502
4	53040061
5	78565534
6	184240878
7	33958497
8	28
10	12

Operand Count Distribution

Operands	Count
0	993048
1	1079574
2	519978803
3	355021753
4	103960248
5	15754261
6	3212311

Register and Memory Operand Counts

- 2 Register Read Operands : 538478789
- 1 Register Write Operand : 684628379
- 3 Memory Operands : 0
- 1 Memory Read Operand : 353933950
- 2 Memory Write Operands : 0

Immediate Values

- Minimum Immediate : -2147483648
- Maximum Immediate : 2147483647

Displacement Values

- Minimum Displacement : -1408
- Maximum Displacement : 135918120

Memory Bytes Touched

- Maximum Bytes Touched : 8
- Average Bytes Touched : 3.7405

3 401.bzip2

Part 1: Dynamic Instruction Profile

Fast-Forward Count : 301000000000

Total Instructions : 302000000002

#	Instruction Type	Count	Percent (%)
1	Loads	452705379	26.885
2	Stores	231174449	13.729
3	NOPs	36514	0.002
4	Direct Calls	791554	0.047
5	Indirect Calls	18	0.000
6	Returns	791569	0.047
7	Unconditional Branches	21299344	1.265
8	Conditional Branches	129923031	7.716
9	Logical Operations	71001117	4.217
10	Rotate & Shift	61832369	3.672
11	Flag Operations	6130	0.000
12	Vector (AVX)	0	0.000
13	Conditional Moves	0	0.000
14	MMX/SSE	0	0.000
15	System Calls	0	0.000
16	Floating Point	0	0.000
17	The Rest	714314267	42.421
Total		1683875741	

Part 2: CPI Analysis

Total Cycles : 48871583873

Total Instructions : 1683875741

CPI : 29.0233

Part 3: Memory Footprint

Unique 32-byte Instruction Chunks : 780

Unique 32-byte Data Chunks : 2528517

Instruction Chunks [single : multi] : 902089947 : 97910045

Data Chunks [single : multi] : 602692921 : 0

Part 4: ia32 ISA Properties

Instruction Length Distribution

Length (Bytes)	Count
1	38611493
2	219201810
3	436966893
4	75326595
5	22046844
6	141356395
7	51341720
8	15085632
10	62610

Operand Count Distribution

Operands	Count
0	36514
1	6147
2	597647608
3	382870395
4	2693762
5	14191359
6	2554207

Register and Memory Operand Counts

- 2 Register Read Operands : 533608733
- 1 Register Write Operand : 712781772
- 3 Memory Operands : 0
- 1 Memory Read Operand : 452705379
- 2 Memory Write Operands : 0

Immediate Values

- Minimum Immediate : -858993459
- Maximum Immediate : 1431655766

Displacement Values

- Minimum Displacement : -4848
- Maximum Displacement : 135000192

Memory Bytes Touched

- Maximum Bytes Touched : 8
- Average Bytes Touched : 3.4854

4 403.gcc

Part 1: Dynamic Instruction Profile

Fast-Forward Count : 107000000000

Total Instructions : 108000000000

#	Instruction Type	Count	Percent (%)
1	Loads	338878362	22.454
2	Stores	170337299	11.286
3	NOPs	3905504	0.259
4	Direct Calls	30223295	2.003
5	Indirect Calls	396083	0.026
6	Returns	30619376	2.029
7	Unconditional Branches	31770287	2.105
8	Conditional Branches	145814338	9.662
9	Logical Operations	54601636	3.618
10	Rotate & Shift	634869	0.042
11	Flag Operations	0	0.000
12	Vector (AVX)	0	0.000
13	Conditional Moves	0	0.000
14	MMX/SSE	0	0.000
15	System Calls	0	0.000
16	Floating Point	0	0.000
17	The Rest	702034612	46.517
Total		1509215661	

Part 2: CPI Analysis

Total Cycles : 36645096270

Total Instructions : 1509215661

CPI : 24.2809

Part 3: Memory Footprint

Unique 32-byte Instruction Chunks : 242

Unique 32-byte Data Chunks : 66105

Instruction Chunks [single : multi] : 952041881 : 47958118

Data Chunks [single : multi] : 504453424 : 0

Part 4: ia32 ISA Properties

Instruction Length Distribution

Length (Bytes)	Count
1	181987982
2	305569875
3	282649273
4	87590742
5	53450476
6	76317809
7	12433842

Operand Count Distribution

Operands	Count
0	3905504
1	2491456
2	426149579
3	363400996
4	173433087
5	30619377

Register and Memory Operand Counts

- 2 Register Read Operands : 482064306
- 1 Register Write Operand : 724663032
- 3 Memory Operands : 0
- 1 Memory Read Operand : 338878362
- 2 Memory Write Operands : 0

Immediate Values

- Minimum Immediate : -2147483587
- Maximum Immediate : 136599840

Displacement Values

- Minimum Displacement : -96
- Maximum Displacement : 138633660

Memory Bytes Touched

- Maximum Bytes Touched : 8
- Average Bytes Touched : 3.7698

5 429.mcf

Part 1: Dynamic Instruction Profile

Fast-Forward Count : 377000000000

Total Instructions : 378000000001

#	Instruction Type	Count	Percent (%)
1	Loads	415211141	27.222
2	Stores	110041876	7.215
3	NOPs	1477682	0.097
4	Direct Calls	12555826	0.823
5	Indirect Calls	0	0.000
6	Returns	12555826	0.823
7	Unconditional Branches	8314617	0.545
8	Conditional Branches	178242665	11.686
9	Logical Operations	75119659	4.925
10	Rotate & Shift	3516304	0.231
11	Flag Operations	0	0.000
12	Vector (AVX)	0	0.000
13	Conditional Moves	0	0.000
14	MMX/SSE	0	0.000
15	System Calls	0	0.000
16	Floating Point	0	0.000
17	The Rest	708217421	46.433
Total		1525253017	

Part 2: CPI Analysis

Total Cycles : 37767711190

Total Instructions : 1525253017

CPI : 24.7616

Part 3: Memory Footprint

Unique 32-byte Instruction Chunks : 65

Unique 32-byte Data Chunks : 11672393

Instruction Chunks [single : multi] : 946575289 : 53424712

Data Chunks [single : multi] : 512816806 : 0

Part 4: ia32 ISA Properties

Instruction Length Distribution

Length (Bytes)	Count
1	80627368
2	485390572
3	315524826
4	50530528
5	22077648
6	5249890
7	40599169

Operand Count Distribution

Operands	Count
0	1477682
2	484787629
3	457346262
4	43832601
5	12555827

Register and Memory Operand Counts

- 2 Register Read Operands : 677524683
- 1 Register Write Operand : 770283518
- 3 Memory Operands : 0
- 1 Memory Read Operand : 415211141
- 2 Memory Write Operands : 0

Immediate Values

- Minimum Immediate : -100000000
- Maximum Immediate : 1374389535

Displacement Values

- Minimum Displacement : -76
- Maximum Displacement : 134957120

Memory Bytes Touched

- Maximum Bytes Touched : 4
- Average Bytes Touched : 4.0000

6 450.soplex

Part 1: Dynamic Instruction Profile

Fast-Forward Count : 364000000000

Total Instructions : 365000000001

#	Instruction Type	Count	Percent (%)
1	Loads	546372489	33.164
2	Stores	101119284	6.138
3	NOPs	11471	0.001
4	Direct Calls	3199444	0.194
5	Indirect Calls	135	0.000
6	Returns	3199579	0.194
7	Unconditional Branches	13157406	0.799
8	Conditional Branches	103115691	6.259
9	Logical Operations	13856059	0.841
10	Rotate & Shift	10329070	0.627
11	Flag Operations	23231804	1.410
12	Vector (AVX)	0	0.000
13	Conditional Moves	0	0.000
14	MMX/SSE	0	0.000
15	System Calls	0	0.000
16	Floating Point	306673197	18.615
17	The Rest	523226144	31.759
Total		1647491773	

Part 2: CPI Analysis

Total Cycles : 46324424110

Total Instructions : 1647491773

CPI : 28.1182

Part 3: Memory Footprint

Unique 32-byte Instruction Chunks : 1177

Unique 32-byte Data Chunks : 5682144

Instruction Chunks [single : multi] : 936838656 : 63161342

Data Chunks [single : multi] : 465313608 : 4659619

Part 4: ia32 ISA Properties

Instruction Length Distribution

Length (Bytes)	Count
1	77086537
2	441651062
3	398975284
4	16506344
5	3277725
6	40943642
7	17620742
8	3938662

Operand Count Distribution

Operands	Count
0	11471
1	23
2	415256457
3	395021811
4	186474439
5	3235797

Register and Memory Operand Counts

- 2 Register Read Operands : 577354595
- 1 Register Write Operand : 764780825
- 3 Memory Operands : 0
- 1 Memory Read Operand : 417091240
- 2 Memory Write Operands : 0

Immediate Values

- Minimum Immediate : -1074790400
- Maximum Immediate : 2147483647

Displacement Values

- Minimum Displacement : -344
- Maximum Displacement : 135856736

Memory Bytes Touched

- Maximum Bytes Touched : 8
- Average Bytes Touched : 5.2526

7 456.hmmmer

Part 1: Dynamic Instruction Profile

Fast-Forward Count : 264000000000

Total Instructions : 265000000004

#	Instruction Type	Count	Percent (%)
1	Loads	547670086	33.737
2	Stores	75698523	4.663
3	NOPs	34317	0.002
4	Direct Calls	144622	0.009
5	Indirect Calls	959	0.000
6	Returns	145581	0.009
7	Unconditional Branches	205862	0.013
8	Conditional Branches	144361645	8.893
9	Logical Operations	1158706	0.071
10	Rotate & Shift	294108	0.018
11	Flag Operations	5669	0.000
12	Vector (AVX)	0	0.000
13	Conditional Moves	0	0.000
14	MMX/SSE	0	0.000
15	System Calls	0	0.000
16	Floating Point	40212	0.002
17	The Rest	853607356	52.583
Total		1623367646	

Part 2: CPI Analysis

Total Cycles : 44635801667

Total Instructions : 1623367646

CPI : 27.4958

Part 3: Memory Footprint

Unique 32-byte Instruction Chunks : 485

Unique 32-byte Data Chunks : 84052

Instruction Chunks [single : multi] : 927512932 : 72487065

Data Chunks [single : multi] : 623212783 : 1000

Part 4: ia32 ISA Properties

Instruction Length Distribution

Length (Bytes)	Count
1	25078720
2	302861340
3	296149985
4	270388611
5	24861161
6	68360378
7	416486
8	11883316

Operand Count Distribution

Operands	Count
0	34887
1	3512
2	566126434
3	432938690
4	713115
5	159551
6	23808

Register and Memory Operand Counts

- 2 Register Read Operands : 562399126
- 1 Register Write Operand : 755280076
- 3 Memory Operands : 0
- 1 Memory Read Operand : 547662203
- 2 Memory Write Operands : 0

Immediate Values

- Minimum Immediate : -987654321
- Maximum Immediate : 2147483647

Displacement Values

- Minimum Displacement : -580
- Maximum Displacement : 135294312

Memory Bytes Touched

- Maximum Bytes Touched : 8
- Average Bytes Touched : 3.9975

8 471.omnetpp

Part 1: Dynamic Instruction Profile

Fast-Forward Count : 43000000000

Total Instructions : 44000000000

#	Instruction Type	Count	Percent (%)
1	Loads	371203494	23.186
2	Stores	229800842	14.354
3	NOPs	802289	0.050
4	Direct Calls	21333587	1.333
5	Indirect Calls	3688428	0.230
6	Returns	25022010	1.563
7	Unconditional Branches	22188322	1.386
8	Conditional Branches	117336435	7.329
9	Logical Operations	60021549	3.749
10	Rotate & Shift	7138155	0.446
11	Flag Operations	20157212	1.259
12	Vector (AVX)	0	0.000
13	Conditional Moves	0	0.000
14	MMX/SSE	0	0.000
15	System Calls	0	0.000
16	Floating Point	96954642	6.056
17	The Rest	625357371	39.060
Total		1601004336	

Part 2: CPI Analysis

Total Cycles : 43070303520

Total Instructions : 1601004336

CPI : 26.9021

Part 3: Memory Footprint

Unique 32-byte Instruction Chunks : 890

Unique 32-byte Data Chunks : 528914

Instruction Chunks [single : multi] : 958077831 : 41922170

Data Chunks [single : multi] : 551087844 : 4472671

Part 4: ia32 ISA Properties

Instruction Length Distribution

Length (Bytes)	Count
1	154604402
2	308590273
3	382033727
4	34345823
5	45116538
6	48807545
7	26501272
10	421

Operand Count Distribution

Operands	Count
0	802289
1	225761
2	518277155
3	281745371
4	172775261
5	25035663
6	1138501

Register and Memory Operand Counts

- 2 Register Read Operands : 540417988
- 1 Register Write Operand : 665526901
- 3 Memory Operands : 0
- 1 Memory Read Operand : 337134926
- 2 Memory Write Operands : 0

Immediate Values

- Minimum Immediate : -2092037281
- Maximum Immediate : 2147483647

Displacement Values

- Minimum Displacement : -104
- Maximum Displacement : 136117612

Memory Bytes Touched

- Maximum Bytes Touched : 8
- Average Bytes Touched : 4.2217

9 483.xalancbmk

Part 1: Dynamic Instruction Profile

Fast-Forward Count : 1331000000000

Total Instructions : 1332000000003

#	Instruction Type	Count	Percent (%)
1	Loads	364640083	23.830
2	Stores	166007418	10.849
3	NOPs	22970337	1.501
4	Direct Calls	12799219	0.836
5	Indirect Calls	8905400	0.582
6	Returns	21702556	1.418
7	Unconditional Branches	8369243	0.547
8	Conditional Branches	180871346	11.820
9	Logical Operations	36973896	2.416
10	Rotate & Shift	5434374	0.355
11	Flag Operations	1615549	0.106
12	Vector (AVX)	0	0.000
13	Conditional Moves	0	0.000
14	MMX/SSE	0	0.000
15	System Calls	0	0.000
16	Floating Point	7056195	0.461
17	The Rest	692843171	45.278
Total		1530188787	

Part 2: CPI Analysis

Total Cycles : 38144866356

Total Instructions : 1530188787

CPI : 24.9282

Part 3: Memory Footprint

Unique 32-byte Instruction Chunks : 2275

Unique 32-byte Data Chunks : 561511

Instruction Chunks [single : multi] : 969411950 : 30588050

Data Chunks [single : multi] : 522262266 : 1899767

Part 4: ia32 ISA Properties

Instruction Length Distribution

Length (Bytes)	Count
1	144654177
2	323626289
3	446361342
4	30948376
5	23407601
6	23173778
7	7455735
8	253090
9	57286
10	62326

Operand Count Distribution

Operands	Count
0	22970337
1	722859
2	420429512
3	416887365
4	99202797
5	23593342
6	16193788

Register and Memory Operand Counts

- 2 Register Read Operands : 456116669
- 1 Register Write Operand : 687997245
- 3 Memory Operands : 0
- 1 Memory Read Operand : 362381635
- 2 Memory Write Operands : 0

Immediate Values

- Minimum Immediate : -1431655765
- Maximum Immediate : 2147483647

Displacement Values

- Minimum Displacement : -1392
- Maximum Displacement : 139657912

Memory Bytes Touched

- Maximum Bytes Touched : 8
- Average Bytes Touched : 4.1663